

CSE4204-8D-T03 Team Information Sheet

Team Name	CSE4204-8D-T03
Section	8D
Project Title	Smart Farmer Assistant: AI-Based Agriculture Support System
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GitHub Repository	https://github.com/sakib-foysal/CSE4204-8D-T03-Smart-Farmer-Assistant

Smart Farmer Assistant – AI-Based Agricultural Support System

Introduction

Agriculture is one of the most important sectors in Bangladesh and many other developing countries. Farmers often face various problems such as plant diseases, unpredictable weather, lack of proper market information, and limited access to agricultural experts. Due to these issues, farmers sometimes suffer financial losses and reduced crop production.

The **Smart Farmer Assistant** is an AI-based mobile application designed to help farmers make better farming decisions. The system will use Artificial Intelligence to detect crop diseases, provide farming suggestions, analyze weather conditions, and recommend suitable actions for better crop management.

This project aims to make modern agricultural technology accessible and affordable for farmers through a smart mobile platform.

Problem Statement

Farmers regularly face challenges that reduce crop quality and productivity. Some major problems include:

- Difficulty identifying crop diseases early
- Lack of instant agricultural guidance
- Unpredictable weather affecting farming decisions
- Limited knowledge about fertilizers and pesticides
- Poor communication with agricultural experts

- Existing systems are not intelligent or personalized

Most current farming applications only provide general information and lack AI-based smart analysis. Farmers need a system that can automatically analyze problems and provide real-time intelligent suggestions.

Artificial Intelligence can help solve these issues by analyzing crop images, weather data, and farming conditions to generate accurate recommendations and predictions.

Objectives

- Build an AI-powered farming assistant mobile application
- Detect crop diseases using image recognition
- Provide smart farming recommendations
- Deliver weather-based farming alerts
- Help farmers improve crop production and reduce losses
- Create an easy-to-use system for rural farmers
- Increase awareness of modern agricultural technology

Proposed Features

The proposed system will contain the following features:

i. User Features

- User registration and login system
- Farmer profile management
- Language support for local farmers

ii. AI-Based Features

- AI crop disease detection using camera images
- Smart fertilizer and pesticide recommendation
- AI chatbot for farming support
- Weather prediction and alerts
- Crop growth monitoring

iii. General Features

- Daily farming tips
- Market price updates
- Notification and alert system
- Dashboard for farming activities
- Expert consultation section

iv. Scope of the Project

The scope of the Smart Farmer Assistant project includes:

- Supporting farmers with intelligent agricultural solutions
- Providing real-time crop disease analysis
- Helping farmers make data-driven decisions
- Offering mobile accessibility for rural users
- Reducing dependency on manual expert consultation
- Improving agricultural productivity through AI

Future improvements may include:

- IoT sensor integration
- Drone monitoring support
- Voice-based AI assistant in Bangla
- Government agriculture support integration

Technology Stack

i. Frontend:

- React.js
- Tailwind CSS
- JavaScript

ii. Backend:

- Django
- Django REST Framework

iii. Database:

- PostgreSQL

iv. AI & Machine Learning:

- Gemini API
- Python
- TensorFlow
- OpenCV

v. Deployment:

- Vercel(Frontend)
- Render(Backend)
- Neon PostgreSQL(Database)

vi. Version Control:

- GitHub

Methodology

The development methodology of this project will follow several stages:

i. Step 1: Requirement Analysis

- Identify farmer problems
- Gather system requirements
- Analyze existing agricultural applications

ii. Step 2: System Design

- Design UI/UX for mobile application
- Create database structure
- Design AI workflow

iii. Step 3: Development

- Build frontend and backend systems
- Develop authentication system
- Integrate APIs and database

iv. Step 4: AI Model Integration

- Train crop disease detection model
- Integrate chatbot and recommendation system
- Test AI prediction accuracy

v. Step 5: Testing

- Functional testing
- User testing
- Performance testing

vi. Step 6: Deployment

- Deploy application
- Monitor system performance
- Collect user feedback

System Milestones

Milestone Tasks		Estimated Time
Phase 1	Requirement analysis and research	Week 1
Phase 2	Minimum Project Structure Build	Week 2
Phase 3	UI/UX design and Database design	Week 3
Phase 4	Frontend and Backend development	Week 4-5
Phase 5	AI model Development and Integration	Week 6
Phase 6	Testing and bug fixing	Week 7
Phase 7	Final Deployment and Polishing	Week 8
Phase 8	Project Show & Presentation	Week 9

Expected Outcome

After completing the project, the system is expected to:

- Help farmers identify crop diseases quickly
- Improve farming productivity and efficiency
- Reduce crop damage and financial losses
- Provide intelligent recommendations using AI
- Increase access to smart agriculture technology
- Improve decision-making for farmers

The AI integration will make the system more accurate, personalized, and efficient compared to traditional farming support systems.

Conclusion

The **Smart Farmer Assistant** is an innovative AI-based agricultural support system designed to solve real-world farming problems. By integrating Artificial Intelligence with mobile technology, the system can assist farmers in disease detection, smart decision-making, and crop management.

This project has the potential to improve agricultural productivity, reduce farmer losses, and promote smart farming practices in Bangladesh and other developing countries. The system will contribute toward building a more technology-driven and sustainable agricultural future.